

3.1. QB - If-else, nested if-else

1. Even or Odd Difference

- Input two integers.
- Print the larger one **only if it is even**; else print the smaller one.

2. Character Classifier

- Input a single character.
- Check if it is **uppercase**, **lowercase**, **digit**, or **special character**.

3. Biggest Among Three

- Input 3 numbers.
- Use nested if-else to find and print the greatest number.

4. Triangle Validity

- Input 3 angles.
- Check if they form a valid triangle (sum = 180 and all > 0).

5. Leap Year Checker

- Input a year.
- Check if it's a leap year using nested conditions (divisible by 4, not 100 unless divisible by 400).

6. Grade Calculator

- Input marks (0–100).
- Print grade based on this:
 - 90–100: A
 - 80–89: B
 - 70–79: C
 - 60–69: D
 - Below 60: F

7. Age Category Checker

- Input age.
- Categorize:
 - 0–12: Child
 - 13–17: Teen
 - 18–59: Adult
 - 60+: Senior

8. Odd Difference Evaluator

- Input two integers.
- If both are odd, print their **absolute difference**. If one is even, print "Invalid".

9. Positive Negative Zero

- Input a number.
- Check if it's positive, negative, or zero.

10. Electricity Bill Calculator

- Input total units consumed.
- Use nested if-else to calculate cost:
 - ≤ 100 units: ₹5/unit
 - 101–300: ₹7/unit
 - 300: ₹10/unit

11. Exam Pass/Fail

- Input marks in 3 subjects.
- If any subject is below 33, print "Fail", else print "Pass".

12. Menu Choice (Math Ops)

- Input two numbers.
- Input a choice (1: add, 2: subtract, 3: multiply, 4: divide).
- Print result using nested if-else.

13. Vowel or Consonant

- Input a character.
- Check if it's a vowel (a, e, i, o, u – case insensitive) or consonant.

14. ATM Withdrawal

- Input balance and withdrawal amount.
- If withdrawal is a multiple of 100 and $\leq$ balance, approve. Else reject.

15. Tax Bracket Checker

- Input income.
- Tax slabs:
 - <2.5L: No tax
 - 2.5L–5L: 5%
 - 5L–10L: 20%
 - 10L: 30%

16. Temperature Converter

- Input a number and a unit (C or F).
- Convert accordingly:
 - If C, convert to Fahrenheit.
 - If F, convert to Celsius.

17. Login Authentication

- Input username and password.
- If both match predefined values (hardcoded), print "Access Granted", else "Denied".

18. Three-Sided Shape Type

- Input 3 sides of triangle.
- Check and classify: equilateral, isosceles, or scalene.

19. Odd/Even Chain Reaction

- Input three numbers.
- If all are odd → print "Odd Chain".

- If all are even → "Even Chain".
- Else → "Mixed Chain".

20. **Simple Voting System**

- Input age and citizenship (Y/N).
- If age ≥ 18 and citizen (Y/y), print "Eligible to vote".

Document by **Suyash** 😊